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# Geophysical evidence for post-Miocene rotation of the island of Viti Levu, Fiji, and its relationship to the tectonic development of the North Fiji Basin

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Geophysical studies of the island of Viti Levu, Fiji, show large amplitude magnetic anomalies of up to 1500 nT to be located above the Tavua volcanic center. Field sampling of the volcanic rocks from the intrusive and extrusive centers of the volcano show the rock samples to have initial intensities of natural remanent magnetization of up to 380 A/m. Radiometric dating of fifteen oriented samples show an age range of 2.9 to 7.0 m.y. B.P. for the suite of rocks collected from the Mba basaltic group. Paleomagnetic analysis of the dated rocks shows declination vectors ranging from $325 \pm 9°$ for an age of $2.9 \pm 0.2$ m.y. B.P. to $290 \pm 4°$ for an age of $7.0 \pm 0.4$ m.y. B.P. As a group, the paleomagnetic declination values of the samples systematically decrease with age. This decrease is attributed to progressive anticlockwise rotation of Viti Levu during the process of formation of the North Fiji Basin. A linear least squares analysis of the paleomagnetic data indicates that the island has rotated in an anticlockwise direction through 90° during the past 7 m.y. at a rate of 13.7° per million years. Declination values as low as 245° measured for some sites (not dated) suggest that anticlockwise rotation of Viti Levu and development of the North Fiji Basin could have commenced as early as 10 m.y. B.P. following the breakup of the ancient Solomons–New Hebrides–Fiji–Lau island arc. Large rotations of microplates may have occurred along other island arcs and marginal basin systems elsewhere in the Western Pacific and such rotational histories may reflect in detail the evolutionary development of these arcs and basins.

## 1. Introduction

Rotational tectonic features such as the spiral system of islands in the Banda Sea Arc of Indonesia and the curved Fiji–Lau Ridge island arc system (Fig. 1) are readily apparent from observations of bathymetric maps. Based on a study of the regional bathymetric maps of the North Fiji Basin [1], Malahoff [2] proposed that the Fijian Islands had undergone anticlockwise rotation (Fig. 2). Further evidence for such a rotation is presented in this paper in the form of initial paleomagnetic and radiometric age results which together show that there has been a progressive rotation of the horizontal paleomagnetic vector (declination) through time. Paleomagnetic results from 15 volcanic sites indicate an anticlockwise rotation of Viti Levu through about 90° during the past 7 m.y.

In another recent study, Malahoff et al. [3] conducted a detailed aeromagnetic survey of the North Fiji Basin. Results from this study suggest the presence of magnetic anomalies 1 to 4 (0–7.8 m.y. B.P.) within the North Fiji Basin and the location of a currently active spreading center to the west of the Fijian Islands. The development of

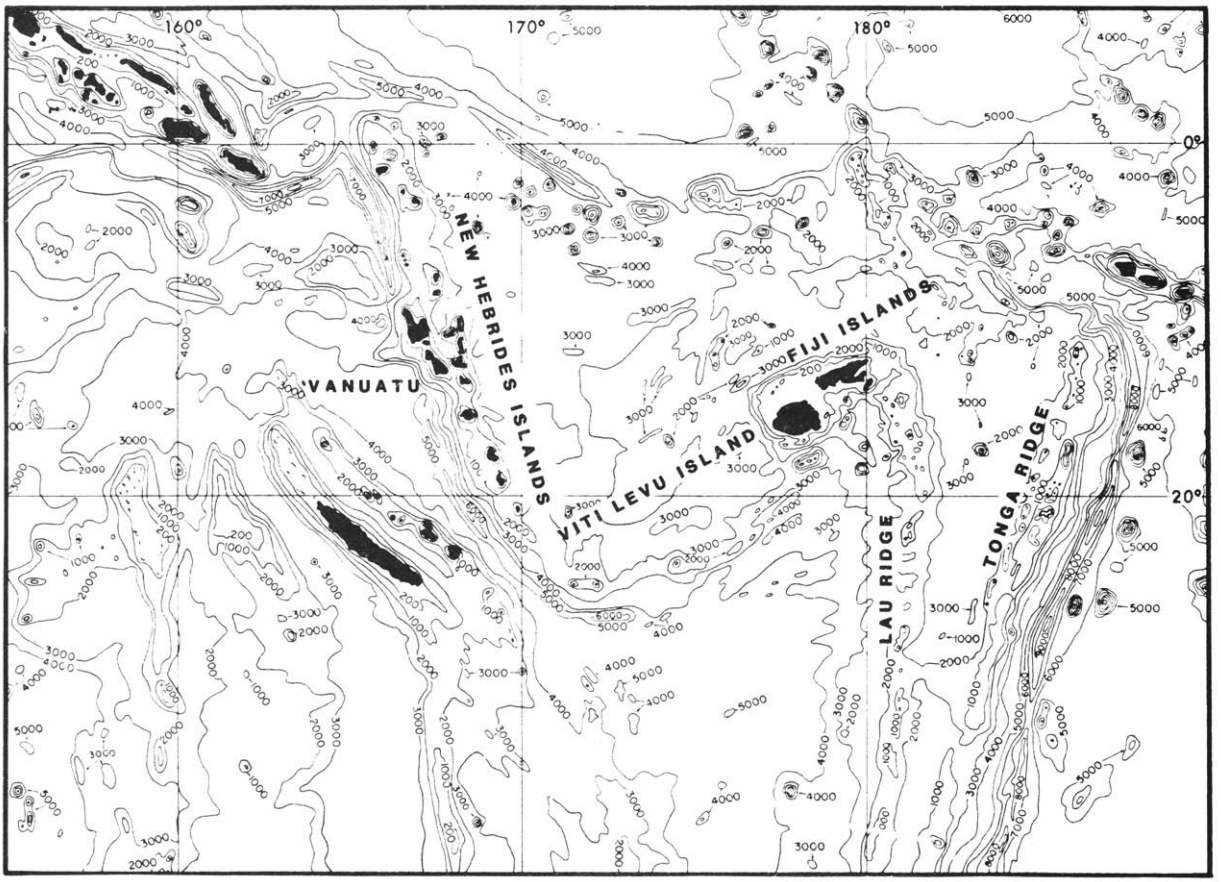



Fig. 1. Simplified bathymetric map of the North Fiji Basin, showing the location of the island of Viti Levu. Water depth in uncorrected meters. Data for the map taken from Mammerickx et al. [1].

a complex pattern of seafloor spreading centers during the past 8 m.y. could have resulted in the Fijian Islands, along with the island of Viti Levu, rotating to their present positions through the process of back-arc spreading, a process similar to that which formed the older South Fiji Basin [4].

## 2. Geological and geophysical background

The North Fiji Basin consists of a complex pattern of bathymetric highs and lows (Fig. 2) associated with an intricate pattern of earthquake foci. Isacks et al. [5] have shown that present subduction of the Pacific plate is taking place under the Tonga–Kermadec Trench, and that transform faulting is probably taking place along the northern boundary of the Tonga Trench where the northwestward moving oceanic crust is being dissected along a hinge-like fault.

Gill and Gorton [6], in studying the geochemical history of the New Hebrides, Fiji and Lau Islands, concluded that the New Hebrides, Fiji, and Tonga Islands formed a continuous ancestral island arc that faced subduction from the northeast. The ancestral island arc underwent contemporaneous volcanism along its entire length during the Early Eocene to Middle Miocene [7]. The arc-trench development and its subsequent migration eastward, toward the Pacific plate, allowed the formation of a back-arc basin, now represented by the North Fiji Basin. During the

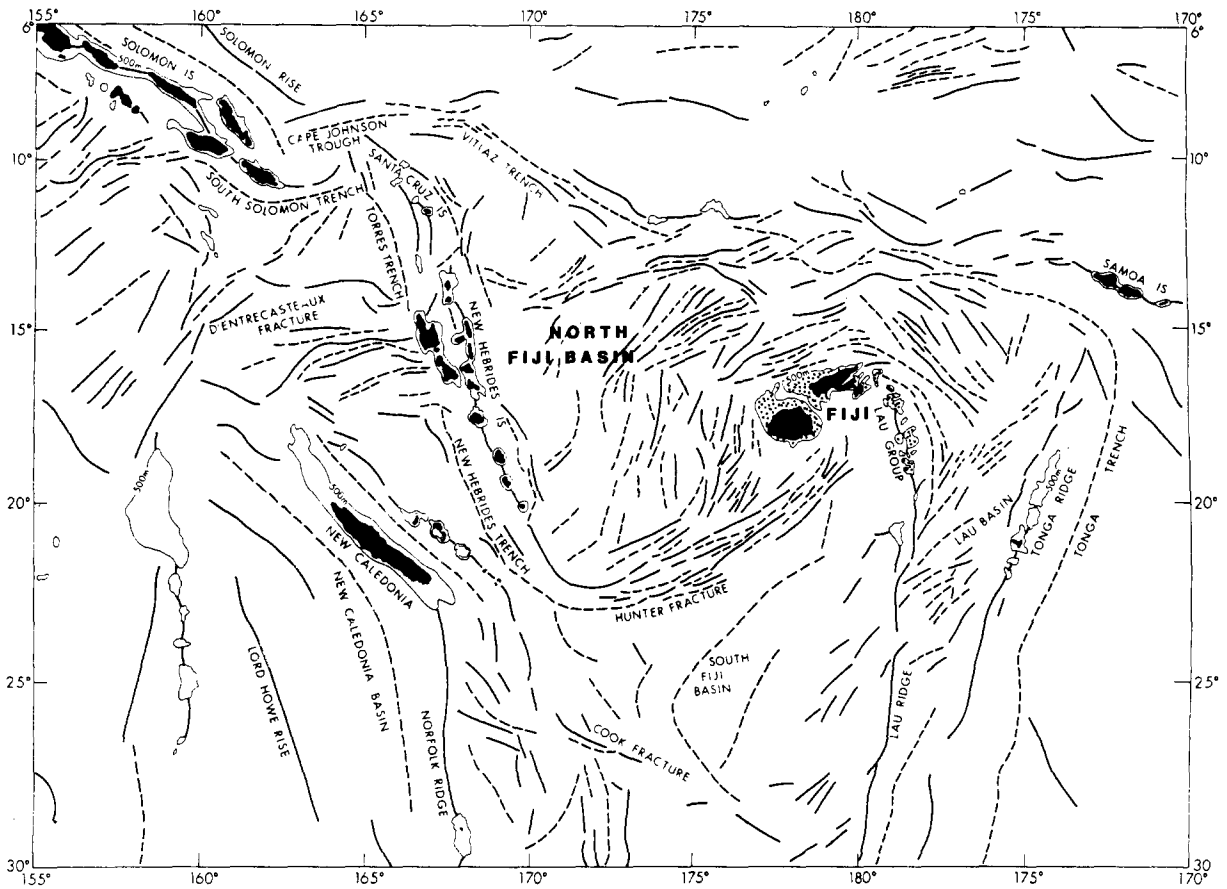



Fig. 2. Bathymetric trends for the North Fiji Basin. Solid lines indicate the location and trend of bathymetric highs, dashed lines indicate bathymetric lows. Data for the trend map taken from Mammerickx et al. [1].

Late Miocene (5.5–9 m.y. B.P.), Gill and Gorton [6] show that a different phase of volcanism commenced between the New Hebrides and Fiji Islands. The Late Miocene was a period of active island arc volcanism with Fiji and the associated subduction zone lying to the north and east of Fiji and the Lau Ridge. At this time the Vitiaz Trench was probably the site of active subduction. During the Pliocene, Fiji ceased to be a typical island arc, while in the New Hebrides, volcanism began to reflect the presence of a subduction zone located to the west. Based upon their geochemical evidence, Gill and Gorton [6] proposed a fragmentation of the New Hebrides–Fiji—Tonga Ridge as most likely beginning between 5.5 and 9 m.y. B.P. although possibly commencing as early as the Middle Miocene. Separation of the Lau from the Tonga Ridge appears to have begun later than 5.5 m.y. B.P. The model for fragmentation of the Fiji–New Hebrides Ridge, as proposed by Gill and Gorton [6], requires an anticlockwise rotation of Fiji beginning between 5.5 and 9 m.y. B.P. as well as the clockwise rotation of the New Hebrides island arc during the same period.

The recently completed aeromagnetic survey of the North Fiji Basin [3] shows the presence of a symmetrical pattern of linear magnetic anomalies located to the west of Viti Levu Island. This sequence is identified as consisting of anomalies 1–4 which implies that the basin was formed between 7.8 m.y. B.P. and the Present. Analysis of this aeromagnetic anomaly sequence suggests that

the North Fiji Basin formed through the process of back-arc spreading during the past 7.8 m.y.

Based on the aeromagnetic evidence [3], the breakup of the New Hebrides–Fiji–Tonga island arc and, hence, the rotation of the Fijian Islands could have commenced at about 7–8 m.y. B.P. The geochemical data of Gill and Gorton [6], which suggest that the separation commenced between 5.5 and 9 m.y. B.P., as stated previously, supports this view.

Geological mapping of the Fijian Islands [9] has shown that the oldest rocks are exposed in the southwest and southern areas of the main island of Viti Levu. These rocks consist of Eocene foraminiferal limestones, as well as andesitic volcanics, and tholeiitic basalt to dacite flows, and volcanoclastics which are intruded by Oligocene tonalite stocks dated 11 to 8 m.y. B.P. [9]. Folding and faulting occurred during a major orogenic period in the Middle and early Late Miocene which predates the proposed Late Miocene Fiji–New Hebrides arc separation [6,7].

During the Late Miocene to Pliocene, the Tavua volcano on Viti Levu (Fig. 3) generated shoshonite-associated rocks; rocks that are typical of island arc volcanism. Geological mapping suggests that the Tavua volcano, which is the largest volcanic center on Viti Levu, ceased activity between 5 and 3 m.y. ago and during its period of activity produced numerous flows and sills, that remain tectonically undisturbed (Fig. 3).

The Tavua volcanic center is also marked as the most prominant geophysical feature on the island of Viti Levu. A Bouguer gravity anomaly of +20 mgal is located over the Tavua volcanic center, implying that the thickness of the crust is at least 30 km beneath the center [10]. The Tavua volcanic center could be regarded as the volcanic root of the island of Viti Levu. Volcanic centers elsewhere, such as those located along the Hawaiian Ridge [11] and the New Hebrides island arc [2], form along distinct crustal rifts, which are, in turn, formed along predominant directions of lithospheric stress. Linear magnetic anomalies associated with such volcanic centers are usually located above dike swarms comprised of highly magnetized rocks [2,11]. Similarly, the Tavua volcanic center is marked by the presence of high-amplitude total magnetic force anomalies of up to 1500 nT peak to peak (Fig. 4), and marks the site of rocks with unusually high intensities of magnetization. The magnetic anomaly map shown in Fig. 4 was constructed from Project Magnet flights [12] carried out over the island of Viti Levu at an altitude of 2200 m. The source of the high-amplitude magnetic anomaly located over the Tavua volcanic center is interpreted to be a highly magnetized volcanic plug with associated dikes, or dike swarms, that may have been intruded as a result of regional tectonic stresses generated during the development of the island. The geological interpretation of the magnetic anomalies, that the volcanic plug and associated dike swarms form the source bodies for magnetic anomalies associated with volcanic centers, has been applied to other volcanic regions as well [2,11,13]. The results of simple depth-to-source analyses of the Tavua magnetic anomalies are illustrated in Fig. 5 and serve to delineate the sites of the principal, possibly dike-filled intrusive centers of Viti Levu Island. These depth-to-source analyses were carried out using the Vacquier "G index" for the inflection points of the observed linear magnetic anomalies [14]. The analyses shown in Fig. 5 suggest that the source bodies for the magnetic anomalies associated with the Tavua volcano radiate away from the Tavua volcanic center. Intensities of natural remanent magnetization (Fig. 6) of rock samples taken from the exposures of rocks from the Mba volcanic group in the vicinity of the Tavua volcanic center agree well with the areal distribution of the total magnetic force anomalies shown in Fig. 5. The Mba Volcanic group is a mapping unit of the geological map of Viti Levu [9] and consists of rocks that are largely Neogene andesites and basalts.

The Tavua volcanic center is therefore characterized by a history of relatively long periods of volcanic activity (Upper Miocene to Pliocene), an apparent lack of post-volcanic tectonic disturbance, and the presence of many exposures of fresh, highly magnetized basaltic and andesitic rocks that are associated with the presence of high-amplitude magnetic anomalies.

Taking the geophysical data into account, together with the earlier suggestions by Tarling [15] that the Tavua volcanic center and rocks from the

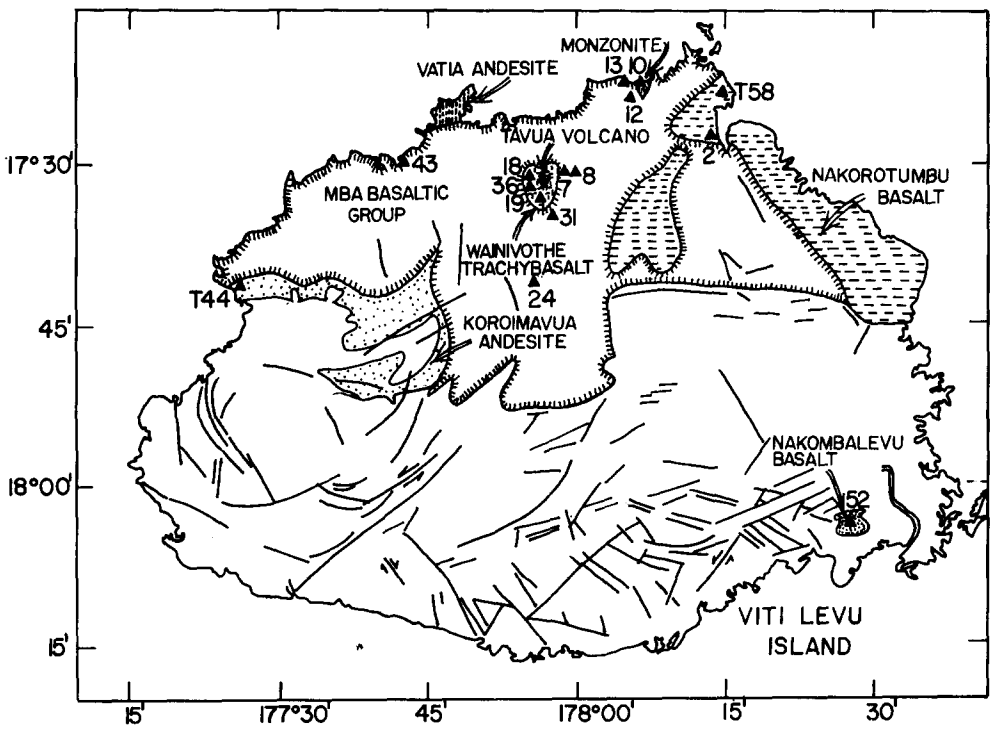



Fig. 3. Principal geological features of the Island of Viti Levu referred to in the text. Geology taken from Rodda and Band [9]. Location of principal faults shown by solid lines. Location and site numbers for rock samples for which radiometric ages were available are shown by solid triangles.

associated Mba volcanic group would likely be the best source for rock specimens for paleomagnetic and radiometric age analyses, the authors concentrated most of their rock sampling efforts within the Mba volcanic group (Fig. 3). Three ranges of intensities of magnetization were encountered in the rock samples, namely those with intensity in the range between 150 and 335 A/m, those with a range between 10 and 15 A/m, and those with a range between 1 and 9 A/m. Rocks having intensities characteristic of the first two groups were sampled from the caldera walls of the Tavua volcano or from exposed dikes. Basaltic rocks sampled from the intrusive or extrusive centers of the Tavua volcano appear to have about one hundred times the initial intensity of remanent magnetization of the rocks taken from exposures of flows outside of intrusive and extrusive centers of the volcano. This analysis of magnetic intensities confirms the deduction from the airborne magnetic data which suggests that the dike rocks of the intrusive centers are the sources of the observed magnetic anomalies observed over the island of Viti Levu.

The dike swarms and the volcanic rocks associated with extrusive phases of the Tavua volcano should contain a history of the rotation of the island of Viti Levu perserved as a systematic rotational offset of paleomagnetic declination vectors with age. Indeed, the earlier sampling by Tarling [15] of volcanic rocks from the Nakorotumbu series, showed anomalous declination vectors of up to 55° west of geographic north (sample T44) for a rock sample radiometrically dated to be 5.35 m.y. years old. Tarling's seemingly anomalous results were another factor which focused the present paleomagnetic study on detailed sampling of the volcanic rocks associated with the Tavua volcanic center. The sample sites together with a simplified geological map of Viti Levu are shown in Fig. 3.

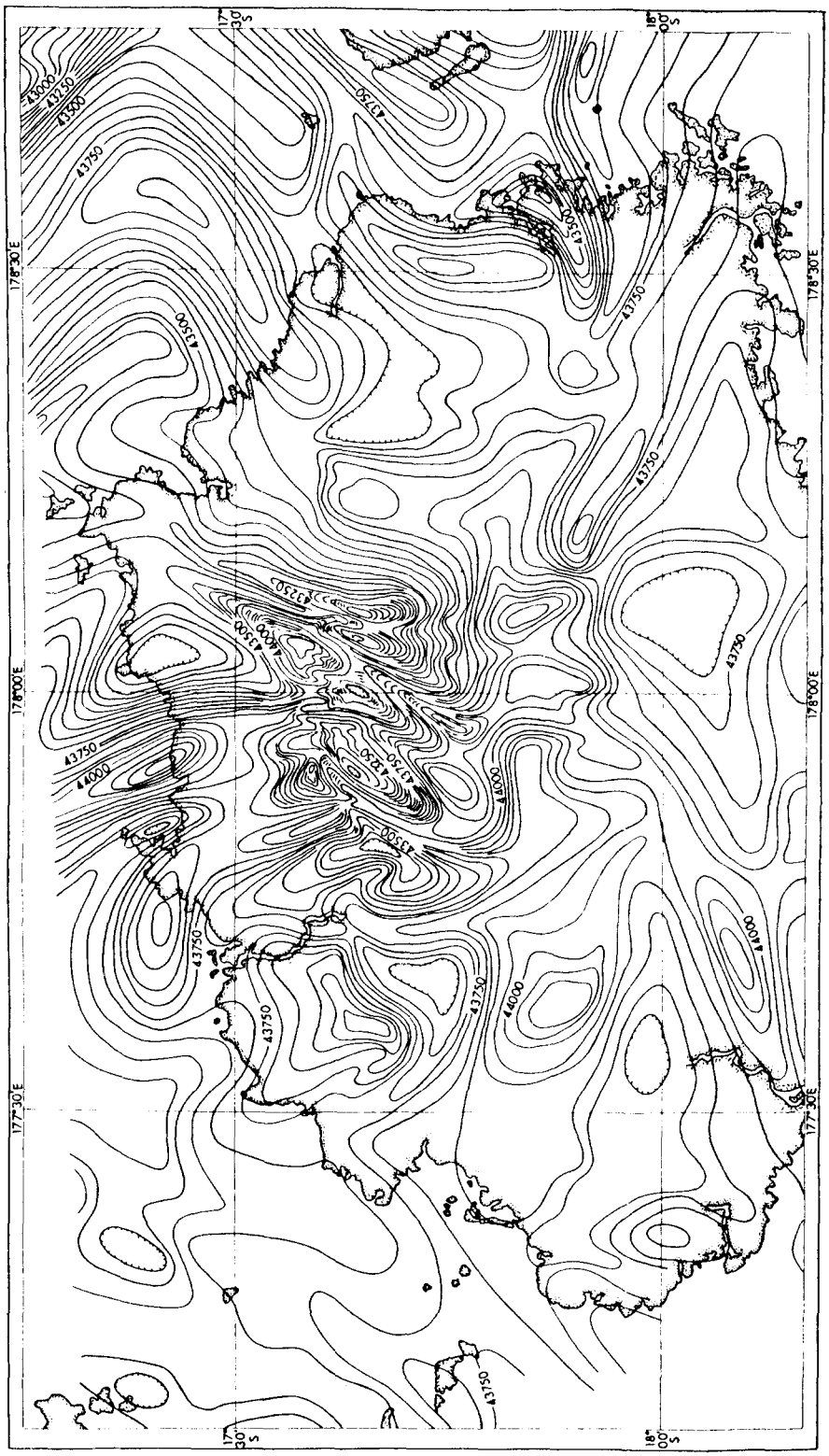



Fig. 4. Airborne total magnetic force anomaly map of Viti Levu Island, based on Project Magnet flight data taken by the U.S. Naval Oceanographic Office [12]. Magnetic intensity values in nanoteslas (nT). Flight altitude was 2200 m above sea level.

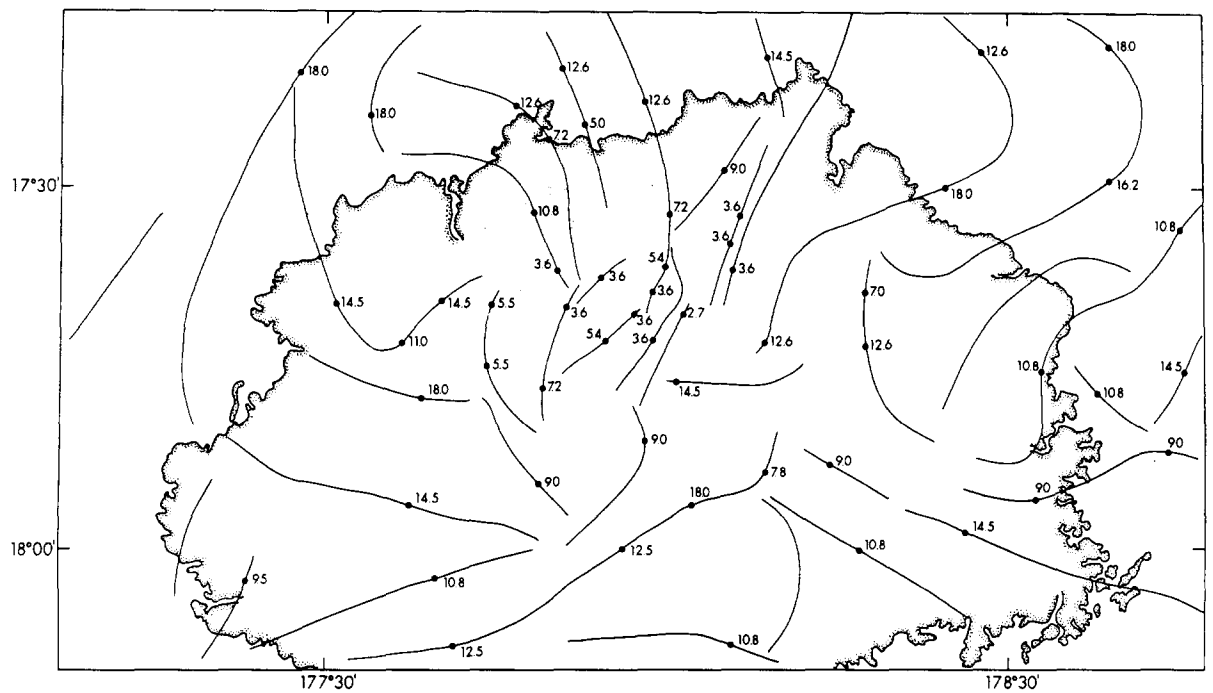



Fig. 5. Location of inflexion lines of bipolar magnetic anomalies for the island of Viti Levu shown in Fig. 4. Estimates of the depths to magnetic source bodies are shown at locations marked by solid circles, in thousands of meters below the flight altitude of 2200 m.

The rocks associated with the Tavua volcanic center are largely andesitic and basaltic flows and intrusions of the Mba, Nakorotumbu, Vatia and Koroimavua volcanic units.

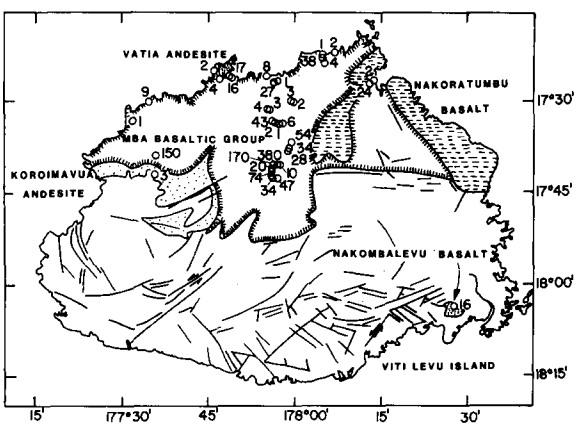



Fig. 6. Location of rock sample sites on Viti Levu, for which initial intensities of magnetization were determined. Intensities of initial magnetization are given in amperes per meter (A/m), geological units taken from Fig. 3.

## 3. Paleomagnetic studies

In order to verify the hypothesis of the anticlockwise rotation [2,6] of the Fijian Islands during the period extending from the Late Miocene to the Present, the authors carried out a field program of sampling tectonically undisturbed volcanic rocks from the island of Viti Levu. The rocks which were ultimately selected for analysis, were chosen on the basis of their suitability for K-Ar dating and stable paleomagnetic properties (Table 1). Rock specimens were collected in the field using a sun compass for orientation and were later drilled and cut into $1'' \times 1''$ cylinders in the laboratory. In most cases two to three samples were collected from each site. Rock exposures with visible signs of weathering or metamorphic alteration were not sampled nor were those for which field observations indicated that they had undergone tectonic disturbance. A total of fifty-two sites were occupied along the northern half of Viti Levu Island (Tables 1, 2, and 3).

TABLE 1

Radiometric age data for the Viti Levu Island rock samples used in this study

| Sample site No. | K [a] (wt.%) | $^{40}Ar_{rad}$ [b] (mol/g × $10^{-12}$) | $\frac{^{40}Ar_{rad} \times 100}{^{40}Ar_{total}}$ (% $^{40}Ar_{rad}$) | K-Ar age [c] (× $10^6$ years) | Rock type [d] |
|---|---|---|---|---|---|
| VL52 | 2.96 ± 0.20 | 14.9 | 18.8 | 2.9 ± 0.2 | olivine basalt |
| VL12 | 1.26 ± 0.05 | 7.74 | 25.1 | 3.7 ± 0.2 | basalt |
| VL10 | 0.802 ± 0.021 | 5.21 | 60.3 | 3.7 ± 1.0 | basaltic dike |
| VL13 | 0.963 ± 0.032 | 6.38 | 32.6 | 3.8 ± 0.2 | trachy-basalt |
| VL7 | 2.25 ± 0.09 | 16.4 | 45.3 | 4.2 ± 0.2 | basaltic dike |
| VL8 | 2.32 ± 0.02 | 35.5 | 37.6 | 4.3 ± 0.5 | basaltic dike |
| VL36 | n.a. | n.a. | n.a. | 4.3 ± 0.1 | monzonite intrusion |
| VL31 | 1.32 ± 0.02 | 10.4 | 24.6 | 4.5 ± 0.2 | basalt flow |
| VL33 | 4.36 ± 0.93 | 35.5 | 34.4 | 4.5 ± 0.2 | basaltic dike |
| VL43 | 2.34 ± 0.05 | n.a. | n.a. | 4.5 ± 0.3 | olivine basalt flow |
| VL19 | 3.06 ± 0.05 | 26.0 | 83.3 | 4.9 ± 0.1 | trachy-basalt |
| VL18 | 1.81 ± 0.06 | 15.5 | 64.4 | 4.9 ± 0.2 | trachy-basalt |
| VL27 | 1.80 ± 0.03 | 16.6 | 39.4 | 5.2 ± 0.1 | basaltic dike |
| VL2 | 0.267 ± 0.005 | 2.43 | 10.3 | 5.2 ± 0.1 | olivine basalt |
| VL24 | 1.48 ± 0.02 | 21.4 | 32.0 | 7.0 ± 0.4 | pillow basalt |

[a] Errors shown are standard deviations of multiple potassium analyses by atomic absorption (AA).

[b] Isotopic analyses of argon in samples were carried out by static mass spectrometry Dalrymple and Lanphere [20].

[c] Constants used: $\lambda_\epsilon = 0.581 \times 10^{-10}$ yr$^{-1}$, $\lambda_\beta = 4.962 \times 10^{-10}$ yr$^{-1}$, $^{40}K/K = 1.167 \times 10^{-4}$ mol/mol. Errors indicated for ages are cumulative standard deviations of the measurements involved in calculating ages.

[d] From field analysis and geological map.

n.a. = geochemical data not available.

Paleomagnetic measurements were made using a Schonstedt DSM-1 spinner magnetometer. The magnetic stability of all samples from each site was evaluated by means of partial alternating-field (AF) demagnetization. Partial demagnetization sequences were carried out using a Schonstedt GSD-1 AF demagnetizer. Each of the samples was subjected to step-wise partial demagnetization to a maximum 600 Oe in order to determine the direction of stable magnetization. Generally, one to five samples from each site were measured (see Table 2). All of the 15 samples which have been radiometrically dated (Table 1) are magnetically stable, having only relatively minor low-coercivity secondary magnetizations which were primarily weak normal axial dipole directions, superimposed on the characteristic magnetization of the reversed polarity samples. The characteristic behavior of the samples during partial demagnetization is illustrated in Fig. 7. Typically, the remanence direction stabilized at about 150 Oe and the declination values shown in Table 1 represent the site mean values for all samples from a particular site at this level.

The mean declination values were obtained from an average of two to five samples from drill cores at each site although at several sites only one useable sample was recovered. Since, a maximum of three flow units were sampled at each site, no attempt was made to account statistically for possible errors due to secular variation on the site coverages (see Table 2). While secular variation could be responsible for some of the scatter indicated in Table 2, a general agreement, in both the sense and amount of rotation, with reconstructions based on the regional magnetic anomalies and bathymetric trends does not appear to be fortuitous and may imply that during the approximately 4-m.y. time interval spanned by the samples, secular variations in the Fiji Island region were relatively small.

All samples were tested for suitability for radiometric dating at the K-Ar dating laboratory at the

TABLE 2

Paleomagnetic data, radiometric age, magnetization, apparent pole positions and geological listings for rock samples from Viti Levu used in this study

| Site No. | Location (lat., long.) | Initial magnetization (A/m) | Number of samples analyzed per site | Paleomagnetic declination [a] $D(°)$ | Paleomagnetic inclination [a], $I(°)$ |
|---|---|---|---|---|---|
| VL52 | 18°03.6'S, 178°27.4'E | 16 | 2 | 325± 9 | 40±7 |
| VL12 | 17°22.7'S, 178°06.6'E | 4 | 1 | 326 | +30 |
| VL10 | 17°22.6'S, 178°06.5'E | 2 | 2 | 339± 3 | +35±10 |
| VL13 | 17°22.4'S, 178°05.0'E | 0.7 | 2 | 319±11 | +20±8 |
| **T58 | 17°22.1'S, 178°15'E | | | 336 | +32 |
| VL7 | 17°31.0'S, 177°59.6'E | 3 | 2 | 329±13 | +27±7 |
| VL8 | 17°31.0'S, 177°59.6'E | 2 | 1 | 328 | 30 |
| VL36 | 17°31.4'S, 177°55.5'E | 3 | 4 | 316±10 | 36±10 |
| VL31 | 17°33.9'S, 177°57.7'E | 6 | 2 | 337± 2 | −45±10 |
| VL33 | 17°33.8'S, 177°57.7'E | 1 | 1 | 323 | −2 |
| VL43 | 17°32.0'S, 177°42.6'E | 3 | 1 | 323 | −16 |
| VL19 | 17°33.2'S, 177°56.9'E | 43 | 3 | 320± 5 | 8±5 |
| VL18 | 17°31.2'S, 177°55.6'E | 4 | 3 | 305± 5 | −30±10 |
| VL27 | 17°32.0'S, 177°54.0'E | 1.5 | 1 | 292 | −34 |
| VL2 | 17°26.8'S, 178°13.9'S | 2 | 1 | 292 | −35 |
| T44 [b] | 17°42'S, 177°28'W | | | 305 | −35 |
| VL24 | 17°41.9'S, 177°56.1'E | 74 | 2 | 290± 4 | 34 |

[a] Declination and inclination are site averages. Declination data adjusted to normal polarity. Plus/minus values for declination and inclination are one standard deviation.

[b] Data from Tarling [15].

Hawaii Institute of Geophysics. The radiometric data are shown in Table 1. The locations of the sample sites are shown in Fig. 3. Samples for which K-Ar dates are not available but which nevertheless underwent paleomagnetic analysis are shown in Table 3.

## 4. Discussion

The North Fiji Basin is one of the youngest currently active marginal basins located in the southwest Pacific. This basin marks the site of at least one active spreading center [2] and the anomaly lineations can be traced back to the time of anomaly 4 (7.8 m.y. B.P.). Gill and Gorton [6], and Gill and McDougall [7] from their geochemical study of the Fiji and New Hebrides volcanic rocks, suggest that the New Hebrides–Fiji–Tonga island arc system existed as one continuous active island arc with a northeastward facing trench (Vitiaz Trench) that existed earlier than 10 m.y. B.P. Furthermore, Gill and McDougall [7] conclude that the change from the Namosi calc-alkaline volcanism to the Koromavua shoshonite volcanism took place somewhere between 5 and 6 m.y. B.P. and suggest that this change resulted from the opening of the Lau Basin. Based on the interpretation of magnetic anomalies, the Lau Basin appears to have opened later at about 4 m.y. B.P., or at the time of magnetic anomaly 3 [8]. The significant change in the chemistry of the Viti Levu volcanism from calc-alkaline andesite to shoshonite as described by Gill and Gorton [6] could have resulted from the breakup of the New Hebrides–Fiji–Tonga island arc and the develop-

| K-Ar age (m.y. B.P.) | Virtual geomagnetic pole positions (lat., long.) | | Geological remarks | Name of formation sampled |
|---|---|---|---|---|
| 2.9 ± 0.2 | 36.7°S, | 39.7°W | olivine basalt from quarry, near Nausori | Mba-Nakombalevu basalt |
| 3.7 ± 0.2 | 42.6°S, | 45.0°W | basalt from bed of Naindandara stream | Mba, undifferentiated |
| 3.7 ± 1.0 | 47.9°S, | 28.4°W | basaltic dike from quarry at Narewa | intrusive monzonite |
| 3.8 ± 0.2 | 40.9°S, | 56.8°W | trachybasalt | Mba, undifferentiated |
| 3.7 | | | basalt | Mba-Nakorotumbu |
| 4.2 ± 0.2 | 45.9°S, | 43.8°W | basaltic dike | Mba, undifferentiated |
| 4.3 ± 0.5 | 43.9°S, | 43.0°W | basaltic dike | Mba, undifferentiated |
| 4.3 ± 0.1 | 32.8°S, | 48.9°W | monzonite intrusion | Mba |
| 4.5 ± 0.2 | 66.9°S, | 114.9°W | flow basalt | Mba, undifferentiated |
| 4.5 ± 0.2 | 50.1°S, | 67.5°W | dike basalt | Mba |
| 4.5 ± 0.3 | 52.8°S, | 77.9°W | olivine basalt flow | Mba |
| 4.9 ± 0.1 | 45.0°S, | 63.1°W | trachybasalt | Mba-Wainivothe trachybasalt |
| 4.9 ± 0.2 | 37.5°S, | 95.1°W | trachybasalt from east of Tavua Caldera on road between Waikumbukumbu and the bridge | Mba-Wainivothe |
| 5.2 ± 0.1 | 25.8°S, | 100.6°W | dike intersections on road | Mba-Wainivothe |
| 5.2 ± 0.1 | 25.9°S, | 101.7°W | olivine basalt from quarry at Nanukuloa | Mba-Nakorotumbu |
| 5.4 ± 0.1 | | | andesite from Lomolomo quarry | Koroimavua andesite group |
| 7.0 ± 0.4 | 12.2°S, | 63.6°W | pillow basalt at head Singatoka River, probable base of Mba volcanic site near village of Nangatangata | Mba, undifferentiated |

ment of rapidly changing and evolving spreading patterns within the young North Fiji Basin.

Tarling's [15] earlier paleomagnetic study of Viti Levu showed that paleomagnetic declination values for samples of Mba basaltic rocks from seventeen sites varied several tens of degrees away from 360°. Although Tarling [15] noted that these directions were magnetically stable and that they did not correspond either to the present geomagnetic field or to expected normal or reversed paleofields, insufficient radiometric age control for his samples precluded any further explanation for these seemingly anomalous observations. Tarling [15] stated, however, that detailed sampling of the Mba volcanic series, together with careful laboratory analysis, would be essential to the solution of the problem.

James and Falvey [16], in a paleomagnetic study of the Suva Marls and two volcanic sites, compared their results with those of Tarling's [15] and developed a hypothesis for 21° of anticlockwise rotation of Viti Levu between the present and 4–5 m.y. B.P. The age of the Suva Marls, however, is not well constrained because of insufficient radiometrically dated sites.

Rocks from the 15 radiometrically dated sites in this paper (Table 1), span an age range of 2.9 to 7.0 m.y. (Table 1). The geological map of Viti Levu [9] was used to locate the sites, and initially, the stratigraphic order of the samples was determined from this map. If only this stratigraphic age had been used in this study, however, errors in the relative age of the rock samples would have resulted. Table 4 shows that an inconsistency exists between the K-Ar ages of the samples used in this study and their stratigraphic position within the

TABLE 3

Paleomagnetic data and calculated age for volcanic rock samples from Viti Levu, for which K-Ar radiometric dates were not available in this study

| Site No. | Location (lat., long.) | | Number of samples analyzed | Initial magnetization (A/m) |
|---|---|---|---|---|
| VL40 | 17°26.3′S, | 177°46.8′E | 1 | 4 |
| VL42 | 17°27.2′ | 177°45.1′ | 3 | 3 |
| VL41 | 17°25.4′ | 177°45.9′ | 2 | 2 |
| VL30 | 17°37.1′ | 177°58.8′ | 2 | 28 |
| VL39 | 17°26.5′ | 177°48.2′ | 3 | 17 |
| VL3 | 17°27.0′ | 178°13.0′ | 3 | 24 |
| VL28 | 17°36.8′ | 177°59.2′ | 2 | 54 |
| VL32 | 17°33.5′ | 177°56.7′ | 1 | 2 |
| VL34 | 17°33.6′ | 177°56.9′ | 1 | 1 |
| VL48 | 17°40.4′ | 177°35.8′ | 4 | 150 |
| VL29 | 17°36.9′ | 177°59.0′ | 1 | 34 |
| VL4 | 17°25.8′ | 177°55.0′ | 3 | 8 |
| VL25 | 17°42.0′ | 177°56.0′ | 3 | 34 |
| VL26 | 17°42.5′ | 177°56.2′ | 3 | 47 |
| VL6 | 17°26.2′ | 177°57.3′ | 2 | 27 |
| VL47 | 17°33.1′ | 177°31.3′ | 2 | 1 |
| VL5 | 17°26.9′ | 177°56.1′ | 2 | 1 |
| VL44 | 17°30.3′ | 177°34.8′ | 3 | 9 |
| VL14 | 17°22.4′ | 178°04.7′ | 2 | 4 |
| VL22 | 17°41.0′ | 177°56.0′ | 2 | 170 |
| VL50 | 17°42.1′ | 177°35.2′ | 3 | 3 |
| VL20 | 17°40.4′ | 177°56.2′ | 1 | 20 |
| VL21 | 17°40.4′ | 177°56.3′ | 3 | 380 |
| VL38 | 17°26.5′ | 177°48.8′ | 1 | 16 |

[a] Plus/minus values for declination and inclination are one standard deviation.

[b] Based on the formula $A=(D-380)/-13.7$ m.y., where 380° is the least squares intercept in Fig. 8; $D$, the paleomagnetic declination of the sample; $-13.7$, the slope of the line in Fig. 8.

volcanic pile of Viti Levu. This inconsistency could have arisen either from errors in the stratigraphic boundaries inherent in a generalized geological map, such as that used in the field phase of this study, or, the errors could have resulted from mislocating the sample sites on the map. Since the sites were ultimately located on detailed topographic maps, the inconsistency between the radiometric and stratigraphic ages, illustrated in Table 3 probably appears as a result of the generalized nature of the geological map of Fiji, from which the formation names for the rock samples used in this study were taken. The radiometric age data suggest that volcanic activity from the Tavua volcanic center extended 7.0 m.y. B.P. (sample VL24) into the Late Miocene.

The paleomagnetic data shown in Tables 2 and 3 confirm the earlier observations of Tarling [15] that declination deviations greater than 30° from true north are preserved within the rocks of Viti Levu. The generally progressive declination deviation west of north for successively older rocks in our data (Table 1) strongly suggest that Viti Levu has been systematically rotated in an anticlockwise direction during the past 7.0 m.y. The oldest rock to have been sampled in our suite (VL24; Table 1 and 2) is from a pillow basalt sequence located near the base of the Tavua volcanic pile. A plot of

| Geological description | Paleomagnetic declination [a], $D$ (°) | Paleomagnetic inclination [a], $I$ (°) | Calculated age [b] (m.y. B.P.) |
|---|---|---|---|
| hornblende andesite—Vatia Series | 358 | −28 | 1.6 |
| olivine augite basalt, undifferentiated Mba | 347 ± 9 | −50 ± 1 | 2.4 ± 0.7 |
| hornblende andesite—Vatia Series | 346 ± 9 | −42 | 2.5 ± 0.7 |
| olivine augite basalts, undifferentiated Mba | 340 ± 2 | −20 ± 2 | 2.9 ± 0.2 |
| olivine augite basalts, undifferentiated Mba | 336 ± 5 | −28 ± 5 | 3.2 ± 0.4 |
| trachybasalts, Lower Mba Series | 330 ± 10 | 53 | 3.7 ± 0.7 |
| olivine augite basalts, undifferentiated Mba | 326 ± 20 | −44 ± 1 | 3.9 ± 1.5 |
| olivine augite basalts, undifferentiated Mba | 325 | −40 | 4.0 |
| olivine augite basalts, undifferentiated Mba | 324 | 24 | 4.1 |
| andesite, Koroimavua Series | 323 ± 6 | −21 ± 6 | 4.2 ± 0.4 |
| olivine augite basalts, undifferentiated Mba | 321 | −46 ± 2 | 4.3 |
| undifferentiated basalts, Mba Series | 320 ± 14 | −24 ± 2 | 4.4 ± 1.0 |
| olivine augite basalts, undifferentiated Mba | 313 ± 15 | −36 ± 1 | 4.9 ± 1.1 |
| olivine augite basalts, undifferentiated Mba | 311 ± 7 | 10 ± 5 | 5.0 ± 0.5 |
| Basalt, Middle Mba Series | 317 ± 10 | 35 ± 2 | 4.6 ± 0.7 |
| olivine augite basalt, undifferentiated Mba | 308 ± 15 | 28 | 5.3 ± 1.1 |
| Trachybasalts, Lower Mba Series | 306 ± 16 | −28 | 5.4 ± 1.2 |
| olivine augite basalt, undifferentiated Mba | 299 ± 9 | 6 ± 4 | 5.9 ± 0.6 |
| trachybasalts, Lower Mba Series | 298 | −25 | 5.9 |
| olivine augite basalts, undifferentiated Mba | 277 ± 12 | 23 | 7.5 ± 0.8 |
| monzonite (Lower Pliocene) | 269 ± 10 | −22 ± 4 | 8.1 ± 0.7 |
| olivine augite basalts, undifferentiated Mba | 266 | −39 | 8.3 |
| olivine augite basalts, undifferentiated Mba | 248 ± 8 | −12 ± 4 | 9.6 ± 0.6 |
| olivine augite basalts, undifferentiated Mba | 245 | −40 | 9.8 |

the paleomagnetic declination data for the rocks listed in Table 2 versus radiometric age, clearly shows a progressive decrease in declination with an increase in the age of the rocks (Fig. 8). These data suggest that the island of Viti Levu, has rotated anticlockwise through a total of about 90° during the past 7.0 m.y. A least squares line fit to the data in Fig. 8 implies that the rotation has been occurring at a rate of nearly 14°/m.y. This average rate appears creditable since, when the limits are applied (±13.6°), the intercept of the line is not greatly different from true north. Our results therefore suggest a considerably greater amount of rotation for Viti Levu than that suggested by James and Falvey [16], i.e., about 90° vs. 21°. Results from two of Tarling's [15] sites, namely T58 and T44, agree well with our data (Fig. 8) and were included in the foregoing analysis. The radiometric age determination by Tarling [15] of 5.35 ± 0.1 m.y. B.P. for samples from site 44 may be regarded as reliable, since Rodda et al. [17] determined ages of 5.3 m.y. B.P. (samples GA459) and 5.4 m.y. B.P. (sample GA460) for the same site. We have assumed that the rocks from Tarling's [15] site T44 are reversely magnetized (based on declination) and, as with the other reversed samples, we have converted them to normal polarity for simplicity of comparison. It must be noted, however, that Tarling's [15] reported inclination value from site T44 is not consistent with the rocks being reversely magnetized and, hence, our interpretation of the results from this site must be

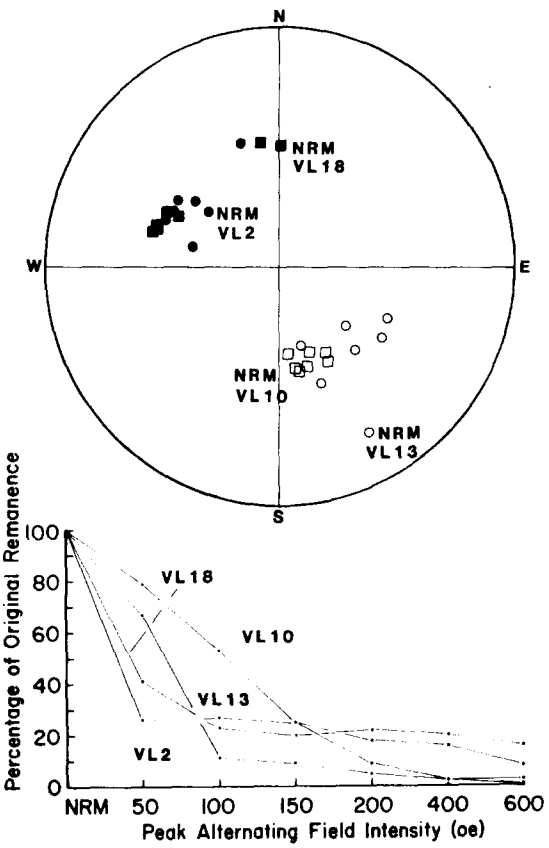



Fig. 7. Characteristic behavior of Viti Levu rock samples VL2, VL10, VL13 and VL18 during partial alternating field demagnetization. Equal-area stereographic projection; closed symbols indicate upper hemisphere, open symbols indicate lower hemisphere.

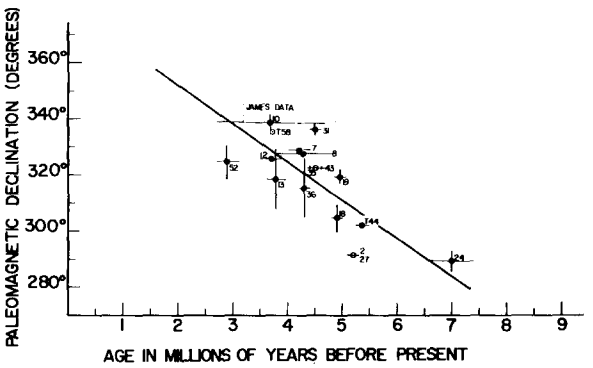



Fig. 8. Site mean paleomagnetic declinations of Viti Levu volcanic rock samples plotted against their K-Ar radiometric ages. All of the declination data for reversed polarity samples have been adjusted to normal polarity. Vertical and horizontal lines indicate the declination and radiometric age errors, respectively. The rectangle marked "James" includes all of James and Falvey's [16] data for Viti Levu. T58 and T44 are Tarling's [15] data points. The least squares fit line is represented by equation $A=(D-380.0)/-13.7$, where $A$ is the radiometric age of the sample, and $D$ is the paleomagnetic declination. James and Falvey's [16] data were not included in the least squares calculations.

regarded as tentative.

If the equation calculated for the least squares line in Fig. 8 is applied to the other samples for which paleomagnetic data are available, but for which no radiometric ages were determined (Table 2), total anticlockwise rotations of greater than 90° and maximum calculated rock ages of about 10 m.y. can be postulated for the sampled volcanic strata. For instance, rock samples VL50, VL20, VL21, and VL38 (Table 3, Fig. 3) show declinations ranging from 269° to 245° and calculated ages ranging from 8.1 to 9.8 m.y. These data and calculations, though tentative, do suggest that the rotation of Viti Levu began as early as about 10 m.y. B.P., that the Mba volcanic group of rocks is perhaps mid-Miocene in age, and that the breakup of the ancient Solomons–New Hebrides–Fiji–Lau island arc began somewhat earlier (about 10 m.y. B.P.) than the 5.5- to 9-m.y. B.P. breakup suggested by Gill and Gorton [6]. The hypothesis of an earlier commencement of volcanism for the Tavua volcano (Mba volcanic group) is substantiated by a radiometric date of $10 \pm 0.5$ m.y. B.P. for samples taken from the Emperor gold mine in the Tavua Caldera by Rodda et al. [17] and for $9.2 \pm 0.5$ m.y. B.P. for samples of the Mba (Vatia) andesite group. Rock samples VL21 and VL38 (Table 2), with the largest deviations from true north, have paleomagnetic declinations of 248° ± 8° and 245° with calculated ages of 9.6 and 9.8 m.y. B.P., respectively. Both sample sites are located within the Mba basaltic group. Sample site VL38 is located within 2 miles of site X57 of Rodda et al. [17]. Rocks from site X57 were radiometrically dated to be $9.2 \pm 2$ m.y. old.

Falvey [18], shows that the New Hebrides island arc rotated 30° in a clockwise direction during the past 6 m.y. Falvey's [18] data and the geochemical data of Gill and Gorton [6] suggest that the clockwise rotation of the New Hebrides island arc commenced 4 m.y. after the initial

TABLE 4

Geological sequence of volcanic formations sampled in this study, together with the K-Ar radiometric dates of the samples [a]

| Geological age | Volcanic group | Rock type | Intrusives | K-Ar age ranges (m.y.) |
|---|---|---|---|---|
| | *MBA basaltic group* | | | |
| | Tova andesite-augite andesite | | | |
| Pleistocene | Nakorotumbu basalt [b] | —basalt flows, conglomerate | | 5.08 ± 0.9, 3.7 |
| | Vatukoro greywacke | —sandstone | | |
| | Rokavukavu basalt | —augite, feldspar basalt | | |
| | Vatia andesite | —hornblende andesite, lava breccia, tuff | | |
| Pliocene | | | | |
| | Tavua trachyandesite [b] | —trachyandesite boulders, tuff, lava | | |
| | | | monzonite intrusives | 3.62 ± 1.05 |
| | Wainivothe trachybasalt [b] | —tuffs, flows, agglomerate | | 4.76 ± 0.17, 4.73 ± 0.1 |
| | Undifferentiated [b] | —olivine basalts | | 6.8 ± 0.44<br>4.48 ± 0.34<br>4.36 ± 0.25<br>4.15 ± 0.55<br>4.06 ± 0.25<br>3.69 ± 0.12<br>3.59 ± 0.15 |
| | Nakombalevu basalt [b] | —olivine basalt | | 2.8 ± 0.2 |
| | *Koroimavua andesitic group* | | | |
| | Undifferentiated [b] | —andesitic flows and pyroclastics | | 5.35 ± 0.1 |

[a] Formation sequence and rock type from "The Geology of Viti Levu" [9].
[b] Formations that have been dated by K-Ar through this study.

breakup of the Solomons–New Hebrides–Fiji island arc that apparently began as early as about 10 m.y. B.P.

The apparent geomagnetic pole positions for Viti Levu, corresponding to the paleomagnetic declinations and inclinations listed for the 15 sites in Table 2 are shown in Fig. 9. The apparent polar wander path for the 15 dated Viti Levu rock samples is strikingly different from that of the New Hebrides island arc polar wander path [18] and the Indonesian polar wander path [19] and is consistent with the view that Viti Levu behaved as a discrete micro-plate.

It is interesting to note that there is a close agreement between the geochemical [5], field magnetic [2,10], and paleomagnetic/radiometric [17] results. These multi-source data together support the hypothesis of the breakup of the Solomons–New Hebrides–Fiji–Lau island arc about 10 m.y. B.P. This breakup resulted in the rotation of the Fijian Islands through at least 90° and perhaps as much as 115° (Fig. 10). The breakup of the ancient island arc probably occurred contemporaneously with the development of spreading centers to the north, west and south of Viti Levu [18].

Large-scale rotations of small plates such as the Fijian Islands could logically be expected in other parts of the world where active plate convergence or marginal basin development is taking place. This would be especially true within broad interaction zones located between two or more major plates, such as the Western Pacific island arc/marginal basin framework located between the Pacific and Indian plates. Rotation of island ridge segments, for example, could be expected in regions of the Western Pacific such as the Banda Island arc area.

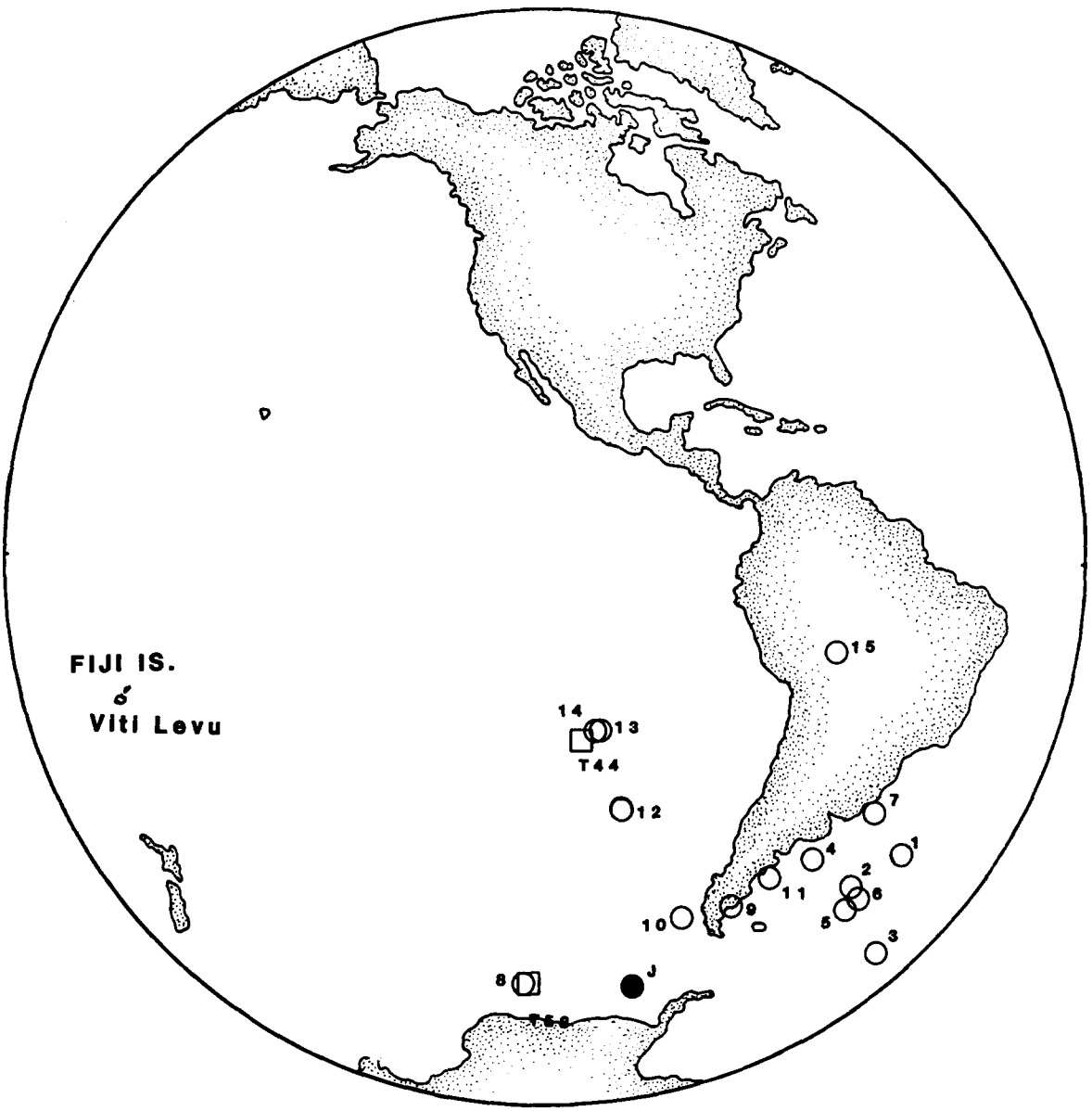



Fig. 9. Virtual geomagnetic pole positions for the rock samples listed in Table 4. The numbers represent the chronological sequence of data shown in Table 4 and Fig. 8, beginning with results from site V152 as number 1. The solid circle is the pole calculated by James and Falvey [16] for the Suva Marls. Squares mark the pole positions corresponding to Tarling's [15] sites 44 and 58. Age (m.y.) of the poles: *1*=2.9±0.2; *2*=3.7±0.2; *3*=3.7±1.0; *4*=3.8±0.2; *5*=4.2±0.2; *6*=4.3±0.5; *7*=4.3±0.1; *8*=4.5±0.2; *9*=4.5±0.2; *10*=4.5 ±0.3; *11*=4.9±0.1; *12*=4.9±0.2; *13*=5.2±0.1; *14*=5.2±0.1; *15*=7.0±0.4.

## 5. Conclusions

(1) The island of Viti Levu, Fiji contains a prominant volcanic center, the Tavua volcano, which was periodically active from about 10 to perhaps 2 m.y. B.P.

(2) Aeromagnetic data show that the Tavua volcano and surrounding rocks of the Mba basaltic

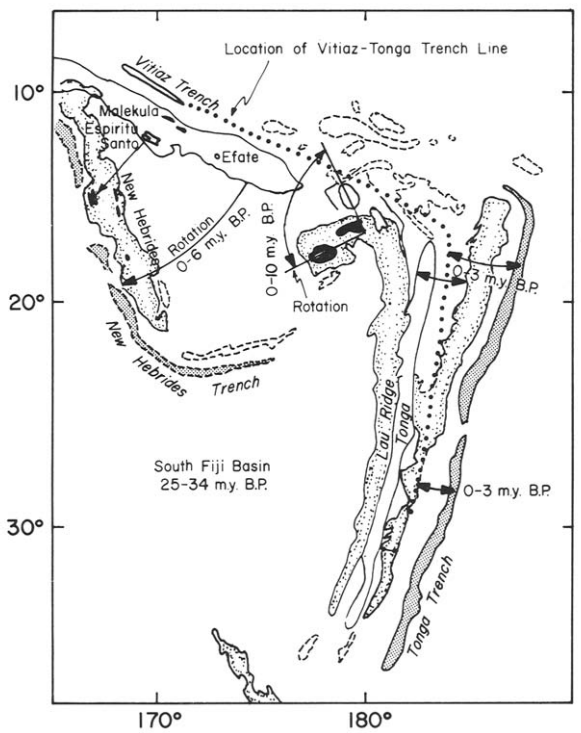



Fig. 10. General schematic illustrating the geological and tectonic evolution of the North Fiji Basin during the past 10 m.y. with the relative positions of the New Hebrides, Fiji, Lau and Tonga island arcs before and after rotation. The illustration also shows the relative eastward advance of the Tonga island arc and trench during this period.

group produce prominant total-force, peak to peak magnetic anomalies of up to 1500 nT.

(3) Detailed sampling of the rocks from the Tavua volcano and of exposures of intrusive and extrusive rocks of the Mba basaltic group, followed by magnetic intensity determinations in the laboratory, show these rocks to be strongly magnetized. Initial natural remanent magnetization intensities are as great as 380 A/m.

(4) The locations of the sampled highly magnetized basaltic rocks may be directly correlated with the position of the observed, high-amplitude total-force magnetic anomalies associated with the Tavua volcano.

(5) Radiometric dating of rock samples from the Tavua volcanic center and the Mba basaltic group collected for this study suggests that Viti Levu underwent periodic volcanism between 2.9 and 7 m.y. B.P. This area of volcanism, located within the northern half of the island, did not undergo apparent significant tectonic disturbance during the post-volcanic period.

(6) Paleomagnetic investigation of thirteen radiometrically dated samples shows paleomagnetic declination vectors to be systematically offset in an anticlockwise direction with increasing age. The oldest rocks dated in the present study have an average paleomagnetic declination of 290° ± 4° corresponding to an age of 7.0 ± 0.4 m.y.

(7) A least squares analysis of the fifteen radiometrically dated samples suggests that the Island of Viti Levu has rotated by about 90° during the past 6.8 m.y., with a rotation rate of 13.7° per million years.

(8) Magnetically stable samples, not radiometrically dated in this study, show declinations as low as 245° for a calculated age of 9.8 m.y. This result suggests that the breakup of the Solomons–New Hebrides–Fiji–Lau island arc and the subsequent rotation of Viti Levu probably commenced as early as about 10 m.y. B.P. A total anticlockwise rotation of about 115° for Viti Levu since the breakup of the ancient island arc is suggested.

(9) Anticlockwise rotation of Viti Levu island was probably generated by the development of spreading centers located northwest and east of the island within the developing North Fiji and Lau marginal basins.

(10) Similar large rotations of other microplates located along the western Pacific island arc and marginal basin boundary with the Indo-Australian plate may be suspected.

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